

Koteswara Rao Madduluri

Senior Android Engineer | Kotlin & Jetpack Compose | Full-Stack (React, Node.js)

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PROFESSIONAL SUMMARY

I'm a Senior Software Engineer with 10+ years of experience, mainly focused on Android (Kotlin/Java) but equally comfortable across the full stack with React, Next.js, and Node.js. Over my career I've worked on 30+ production apps across loyalty and fintech, digital banking, smart city IoT, EdTech, and enterprise software. I've led Jetpack Compose migrations across hundreds of screens, driven multi-module restructuring that cut build times by around 30%, and managed offshore teams of 4-5 engineers while working closely with stakeholders and clients. I hold a Backbase Certified Android Engineer certification and have deep hands-on experience with payment terminal integrations, EPOS systems, AI/ML on device, and real-time communication. I care about writing code that's clean, maintainable, and actually works in the real world.

KEY SKILLS

Mobile: Kotlin, Java, Jetpack Compose, XML Layouts, Kotlin Multiplatform (KMP), Compose Multiplatform (CMP), Kotlin Coroutines, Flow, StateFlow

Architecture: MVVM, MVI, Clean Architecture, Multi-Module, Modular Design

Android Libraries: Retrofit, Ktor, Volley, Koin, Hilt, Dagger, Room, WorkManager, Navigation Component, ExoPlayer, CameraX, Coil, Glide

Frontend: React, Next.js, TypeScript, JavaScript, HTML, CSS, SCSS

Backend: Node.js, NestJS, Express, REST API Development

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase Realtime DB, Room, Datastore

Realtime & IoT: MQTT, XMPP, WebRTC, Socket.IO, BLE, NFC, Nano HTTP

Firebase: Auth, Firestore, FCM, Crashlytics, Remote Config, App Distribution

AI / ML: TensorFlow, TensorFlow Lite, ML Kit

AI-Assisted Development: Claude Code, Cursor, ChatGPT (Codex)

Testing: JUnit, Espresso, Robolectric, MockK

DevOps & CI/CD: GitHub Actions, Bitrise, Play Console, Firebase App Distribution

Security: R8/ProGuard, SSL Certificate Pinning, Play Integrity API

Integrations: Verifone, PAX, Castles, Ingenico, Stripe, Glic, Zelle, ShuftiPro, Payveris, Threat Metrix

Tools: Android Studio Profiler, LeakCanary, Sentry, Postman, Swagger, pgAdmin, VS Code, IntelliJ, Gradle (Kotlin DSL), Linear

Version Control: GitHub, GitLab, SVN

Methodology: Agile, Scrum, Sprint Planning, Code Review, Mentoring

PROFESSIONAL EXPERIENCE

Senior Android & Full Stack Engineer | Lopay

Feb 2026 to Present

Projects: Terminal Pay, Lopay - Point of Sale (EPOS) Products

- Working across two POS products - Terminal Pay and Lopay - building out the EPOS modules on Android. Both products run on dual-screen devices where the customer display and operator interface need to stay in sync, which adds an interesting layer of complexity.
- On Terminal Pay I own the product end to end on my own, across every area - Android, iOS and the backend.

- Leading the migration of both apps from XML to pure Jetpack Compose - we're talking hundreds of screens across the two products. It's a big undertaking but the difference in developer experience and UI consistency is already noticeable.
- Restructuring both codebases from monolithic single-module setups into proper multi-module architectures. This made it much easier for the team to work in parallel without stepping on each other.
- Putting MVVM in place across all modules - keeping the code clean, testable, and easy to reason about day to day.
- Building and maintaining backend APIs in NestJS to support the POS features, and jumping into the React web app when needed. It's genuinely a full-stack role. I also work with the Stripe SDKs and APIs across the payment flows.
- Built proof-of-concept apps with Kotlin Multiplatform (KMP) and Compose Multiplatform (CMP) to evaluate sharing business logic and UI across platforms for future Lopay products.
- Actively use AI tooling - Claude Code, Cursor, and ChatGPT (Codex) - as part of the day-to-day development workflow, speeding up feature delivery, refactoring, and test coverage.
- Working directly with stakeholders to understand what's needed and turn it into something the team can actually build and ship.

Senior Android & Full Stack Engineer | [Payment Loyalty](#)

Nov 2023 to Jan 2026

Projects: *Payment Loyalty Platform, Lopay (May 2025 to Jan 2026)*

- Owned the complete Payment Loyalty solution on my own, end to end. The system I inherited was old and faulty, so I restructured it from the ground up. It has been stable ever since with zero downtime.

Worked on a loyalty platform that sits alongside payment terminals - Verifone, Ingenico, PAX, Castles. When a customer pays, the loyalty flow kicks in between the transaction. It sounds simple but getting it to work reliably across different hardware in live retail environments takes a lot of careful engineering.

- Migrated 40-60 screens from XML to Jetpack Compose and built a custom design system from scratch - reusable components that could be shared across multiple apps and modules without things breaking.
- Broke the app apart from a single module into a proper multi-module architecture organised by feature and core layers. Build times dropped by around 30% and the codebase became a lot easier to navigate and maintain.
- Built and maintained backend APIs in Node.js and NestJS, and regularly contributed to the React web app too. It wasn't just a mobile role - I was involved across the whole stack.
- Led a small offshore team of 4-5 engineers. Used Linear to manage tasks, ran code reviews, kept things moving across time zones, and handled the coordination with third-party vendors like Verifone, PAX, and Stripe.
- Set up WorkManager and background Services to keep data syncing reliably even when the app was offline - important for a platform running in shops and restaurants all day.
- Integrated Firebase Crashlytics and Sentry so we could catch crashes and ANRs quickly. Made a real difference to how fast we could diagnose and fix production issues.
- Did regular performance profiling with Android Studio Profiler and LeakCanary - tracked down memory leaks and fixed a few things that were quietly slowing the app down.

Senior Software Engineer | [Mobiquity \(Hexaware\)](#)

Aug 2021 to Nov 2023

Projects: *Sandy Spring Bank, CNB Bank - Android Mobile Banking Apps (Backbase Digital Banking Platform)*

- Worked on the Backbase digital banking platform for two US banks - Sandy Spring Bank and CNB Bank. Delivered features across Accounts, Loans, Payments, Profile, and Remote Deposit Capture.
- Acted as the Lead Engineer and main point of contact for the delivery team. Set the goals, kept things on track, unblocked the team when needed, and ran knowledge transfer sessions when new people came on board.
- Integrated SDKs like Threat Metrix, Glia, Zelle, and Payveris into the Backbase product - tricky work because you're fitting third-party tools into an existing architecture without breaking what's already there.
- Built shared components used across multiple teams and kept a close eye on code quality through reviews and pair programming sessions.
- Implemented SSL certificate pinning and other security measures - working to banking-grade standards means you can't cut corners on this stuff.
- Kept the CI/CD pipelines running smoothly on GitHub Actions and App Center, and managed Play Store releases across dev, staging, and production environments.
- Worked in 2-week Agile sprints with daily standups. Also ran POC and SPIKE investigations when we needed to evaluate a new approach before committing to it.

- Used Kotlin Coroutines, Flow, and the repository pattern throughout - all within a clean architecture that made the code easy to test and maintain.

Senior Software Engineer | Sensen Networks

Oct 2020 to Aug 2021

Projects: *Gemineye - Smart City IoT Platform*

- Used TensorFlow, TensorFlow Lite, and ML Kit to build vehicle number plate detection and recognition into the platform. Reduced a lot of manual processing and improved the accuracy of the system significantly.
- Rewrote the communication layer between the client and admin apps using NanoHTTP SDK and Content Providers - the old approach wasn't reliable enough for a live IoT environment.
- Migrated the networking layer from Volley to Retrofit, which made the code cleaner and a lot easier to maintain going forward.
- Used Kotlin Coroutines and LiveData throughout to handle async operations properly and keep the app responsive - ANRs dropped noticeably after this.
- Built reusable UI components shared across multiple modules and worked closely with the UI/UX team on some of the more complex screen layouts.
- Ran pair programming sessions and code reviews with junior developers - I find that's the best way to raise the quality of the whole team, not just your own work.

Software Engineer | Iconma

Oct 2017 to Oct 2020

Projects: *Enterprise Collaboration & EdTech Applications*

- Built and maintained enterprise collaboration and EdTech apps for a range of clients, covering the full Android development lifecycle from requirements through to release.
- Led the migration of several core apps from Java to Kotlin - cleaner code, less boilerplate, and it opened the door to using modern Android libraries properly.
- Built real-time chat features using XMPP and Socket.IO and developed the backend APIs that powered the messaging and collaboration functionality.
- Applied design patterns like Singleton, Builder, Factory, and Observer consistently - and kept SOLID principles front of mind throughout.

Software Engineer | Tvisha Technologies

Jul 2015 to Sep 2017

Projects: *Travel Planning, Real-Time Communication & Payment Gateway Apps*

- Started my career here building mobile and web apps across travel, real-time communication, and payments - good grounding across a variety of different problem spaces.
- Got solid foundations in REST API integration, payment gateway implementation, and building Android UIs to Material Design standards.

EDUCATION & CERTIFICATIONS

Backbase Certified Android Engineer - Backbase

2023

B.Tech in Computer Science - Jawaharlal Nehru Technological University, India

2015